

Overview

Non-functional targets: 10k orders/day, p99 < 200ms, 99.9% uptime. Key patterns: CQRS, Kafka event streaming, API Gateway, consistent hashing, distributed tracing, and structured observability.

Capacity Estimates

Metric	Value	Notes
Orders / day	10,000	~0.12 orders/second average
Peak QPS (10× avg)	~1.2 orders/sec	Flash sales may spike higher
p99 latency target	< 200 ms	End-to-end API response
Availability	99.9%	≤ 8.7 hrs downtime/year
DB writes / day	~50k rows	Orders + items + audit log

CQRS Split



Kafka Topics

Topic	Producer	Consumer(s)	Purpose
orders.placed	OrderEventProducer	InventoryAlertConsumer, NotificationConsumer	Context order event
orders.shipped	OrderEventProducer	NotificationConsumer	Shipping notification
inventory.low	InventoryService	NotificationConsumer	Restock alert
search.sync	ProductService	SearchSyncConsumer → Elasticsearch	Search index updates
orders.placed.DLT	Kafka (auto)	DLT handler	Dead letter for failed processing

Observability Stack

Pillar	Tool	ShopVerse Integration
Metrics	Micrometer → Prometheus → Grafana	OrderPlacementTimer, FailedPaymentCounter, ActiveCartGauge
Tracing	OpenTelemetry → Jaeger (dev) / Zipkin (prod)	Auto-instruments HTTP, JDBC, Redis, Kafka; sampling=1.0 in dev
Logging	Logback + logstash-logback-encoder	JSON lines with traceId, spanId, customerId, orderId MDC fields
Correlation ID	CorrelationIdFilter	X-Correlation-ID header → MDC → response